CSCI218 Roundup

SAQ

Topic: Search (2 Questions)

*1. [A Expansion Calculation]**

Consider a graph where Start node S has two children A and B.

- $g(S) = 0, h(S) = 5$
- $c(S, A) = 2, h(A) = 3$
- $c(S, B) = 4, h(B) = 1$
- Goal G is a child of A with $c(A, G) = 2, h(G) = 0$.

  (a) Calculate the f-value ($f(n) = g(n) + h(n)$) for nodes S, A, B.

  (b) Which node does A* expand first after S?

2. [Heuristic Admissibility Check]

You are solving a pathfinding problem where the true optimal cost from node X to Goal is $h^*(X) = 10$.

- Heuristic $h_1(X) = 8$
- Heuristic $h_2(X) = 12$
- Heuristic $h_3(X) = 10$

  (a) Which of these heuristics are admissible?

  (b) Which heuristic is most informative (dominates the others) while remaining admissible?

Topic: Knowledge & Logic (2 Questions)

3. [Entailment via Model Checking]

Consider a knowledge base with two propositions P and Q.

$KB = \{P \to Q, P\}$

(a) Construct a truth table for P and Q.

(b) Identify the rows where the KB is true.

(c) Does $KB \models Q$? Explain based on the rows identified.

4. [Resolution Refutation Step]

You want to prove R using Proof by Contradiction.

Clauses: 1. $\neg P \vee Q$, 2. $\neg Q \vee R$, 3. P, 4. $\neg R$ (Negated Goal).

(a) Resolve Clause 3 (P) and Clause 1 ($\neg P \vee Q$). What is the resolvent?

(b) Resolve the result from (a) with Clause 2. What is the new resolvent?

(c) Does this lead to a contradiction with Clause 4?

Topic: Probabilistic Reasoning (2 Questions)

5. [Bayes Rule Calculation]

A medical test for a disease has the following properties:

- Prevalence $P(D) = 0.01$
- Sensitivity $P(+|D) = 0.95$
- False Positive Rate $P(+|\neg D) = 0.05$

  (a) Calculate the numerator $P(+|D)P(D)$.

  (b) Calculate the denominator $P(+)$ using total probability.

  (c) Compute $P(D|+)$.

6. [HMM Joint Probability]

Consider an HMM with State $X_t \in \{Rain, Sun\}$ and Evidence $E_t \in \{True, False\}$.

- $P(X_0 = Rain) = 0.5$
- $P(X_1 = Rain|X_0 = Rain) = 0.7$
- $P(E_1 = True|X_1 = Rain) = 0.9$

  (a) Write the formula for $P(X_0 = Rain, X_1 = Rain, E_1 = True)$.

  (b) Calculate the numerical value.

Topic: ML & Computer Vision (4 Questions)

7. [Perceptron Weight Update]

Given perceptron weights $w = [-2, 3, 1]$ ($w_0 = -2$ is bias), input $x = [1, 2, 1]$ ($x_0 = 1$ for bias), true label $y = -1$, and learning rate $\alpha = 0.5$.

(a) Compute the prediction $\hat{y} = \text{sign}(w \cdot x)$.

(b) Apply the update rule $w \leftarrow w + \alpha \cdot y \cdot x$. Show the new weight vector.

8. [Convolution Output Size]

Given a 5×5 input image, 3×3 filter, stride = 1, no padding.

(a) Calculate the output feature map dimensions using the formula: $\lfloor (I - F + 2P)/S \rfloor + 1$.

(b) If we want the output to remain 5×5, what padding P is required?

9. [L2 Regularization Gradient]

The loss function with L2 regularization is $L(w) = \text{Error}(w) + \lambda \sum w_i^2$.

(a) What is the derivative of the regularization term λw_i^2 with respect to w_i?

(b) How does this term affect the weight update rule $w_i \leftarrow w_i - \eta \frac{\partial L}{\partial w_i}$?

10. [Metric Threshold Scenario]

You have a classifier outputting probabilities. Current Threshold = 0.5.

- At 0.5: Precision = 0.9, Recall = 0.4.
- You need to improve Recall for a safety-critical application.

  (a) Should you increase or decrease the threshold?

  (b) What will likely happen to Precision as a result?

Topic: NLP (2 Questions)

11. [N-gram Probability Calculation]

Consider a Bigram Model.

- Count("the") = 1000
- Count("the cat") = 50

  (a) Write the formula for $P(\text{cat}|\text{the})$.

  (b) Calculate the probability.

  (c) What problem arises if Count("the cat") = 0?

12. [Embedding Similarity]

You have two word vectors:

- $v_{\text{king}} = [1, 0, 0]$
- $v_{\text{queen}} = [0.9, 0.1, 0]$

  (a) Calculate the Dot Product $v_{\text{king}} \cdot v_{\text{queen}}$.

  (b) Are these words semantically similar based on this score?

  (c) How would this compare to One-Hot vectors for "king" and "queen"?